What Explains How Long the Power Stays Out?

County Outage Duration across Nine U.S. Hurricanes

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Wind explains most of which counties stay without power longest in a hurricane. A model that has never seen a storm ranks its counties by outage duration with a rank correlation of 0.77. Ranking them by gust alone already gives 0.71. With boosted trees, local characteristics add 0.03 and rain 0.01; only the first clears my threshold, and neither does with a linear model. My radar flood measure adds nothing, and satellite foundation-model embeddings can stand in for hand-built local variables but do not add to them.

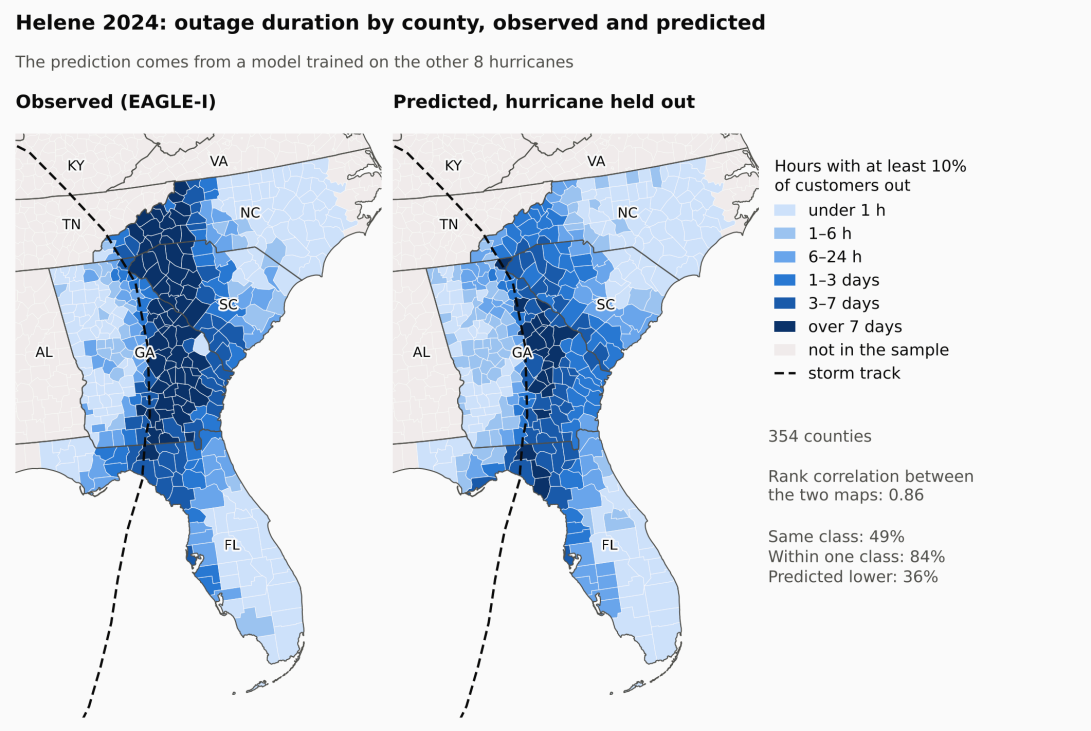


Figure 1. Helene 2024, predicted by a model trained on the other eight hurricanes. The model finds the corridor east of the track but underestimates the longest outages: 68 counties were out for more than 7 days, and it predicted 21.

Question and data

Which public data explain how long a county stays without power, when the model is tested on a hurricane it has not seen?

- **Outcome**: hours with at least 10% of a county's customers out, from EAGLE-I, the 15-minute outage record of Oak Ridge National Laboratory.
- **Sample**: 1,767 county-hurricane pairs in 617 counties and 8 states, from nine hurricanes between Irma (2017) and Helene (2024). Eleven were screened, and Harvey and Sally had too few long outages on record. A county enters when it lies inside the 34-knot wind radius or receives 100 mm of rain in 3 days. In 620 pairs the outage reached 24 hours, and in 756 it never reached 10%.
- **Predictors**: wind (8 variables, from RTMA gusts and the IBTrACS track), rain (3, PRISM), place (20: land cover, terrain, census), flood (10: Sentinel-1 flood extent, land close to stream level) and AlphaEarth satellite embeddings (64 dimensions, reduced to 16).

How I tested it

Every score comes from a hurricane the model did not train on. I train gradient-boosted trees on eight hurricanes, predict the ninth, and repeat nine times. The score is the Spearman rank correlation between predicted and observed duration inside the held-out hurricane, averaged over the nine.

- **A rule set before any result.** A group of variables helps if the mean score rises by 0.02 and at least 6 of 9 hurricanes improve.
- **A strict test, added afterwards.** It also removes the held-out hurricane's counties from training. It trains on less data as well, so a lower score has two possible causes.

The model explains outages after the fact: it uses observed wind and rain, not forecasts.

What I found

Wind dominates the ranking. Gust alone scores 0.71, a wind-only model 0.73, wind, rain and place 0.77, and all variables 0.79. The 0.77 runs from 0.65 to 0.89 across hurricanes. Shuffling the wind variables in the full model costs 0.66, and shuffling any other group at most 0.03.

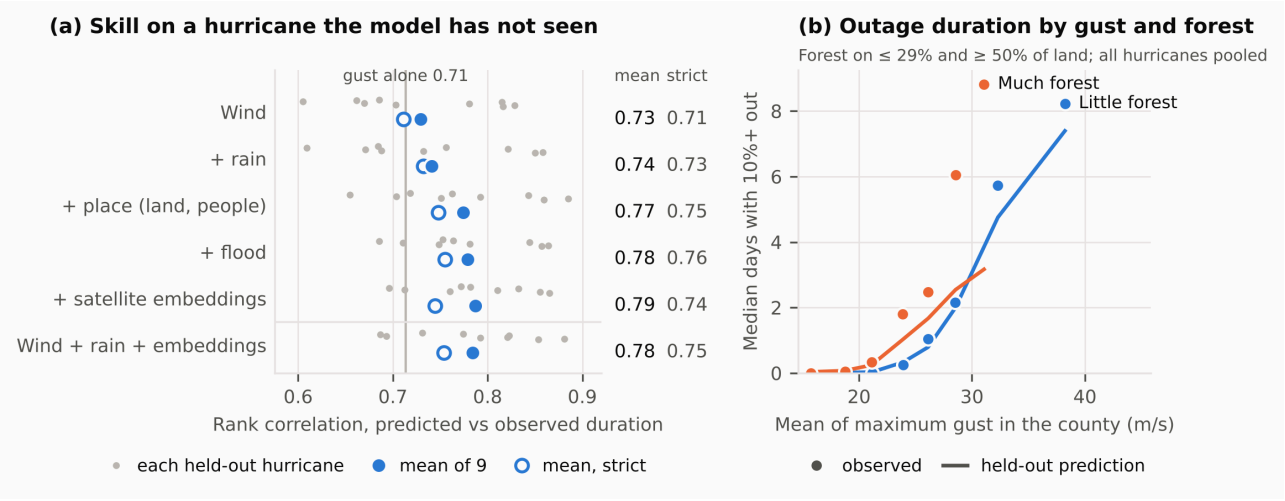


Figure 2. (a) Score of each set of variables on unseen hurricanes. (b) Median outage duration by gust for the least and most forested thirds of counties, all hurricanes pooled.

Only place passes my rule, and not robustly.

Group added: change in mean score (hurricanes that improve)	Standard test	Strict test
Rain, to wind	+0.012 (7 of 9)	+0.021 (8 of 9)
Place, to wind and rain	+0.033 (9 of 9)	+0.016 (7 of 9)
Flood, to all of the above	+0.004 (5 of 9)	+0.007 (6 of 9)
Embeddings, to all of the above	+0.009 (7 of 9)	−0.011 (3 of 9)

Place clears the bar on the standard test and falls below it on the strict one. Rain does the reverse, by a margin that disappears with another library version (+0.016). With a ridge regression, no group beyond wind passes. One pattern behind place: at gusts of 27.5 to 30 m/s, the most forested third was out for a median of 145 hours, against 52 for the least forested third. Half of those forested pairs come from Helene, so forest and storm are hard to separate.

My radar flood measure adds no skill. It also failed its own check: flooded share and 3-day rain correlate at only 0.01, or 0.07 with cropland excluded. Only 61% of pairs had a Sentinel-1 image within 6 days, and cropland made up 39% to 72% of the flagged area in a typical county. This says little about flooding itself.

Embeddings can stand in for hand-built place variables but do not add to them. Wind, rain and embeddings score 0.78, against 0.77 with the 20 place variables, and 0.75 for both on the strict test. Added on top, embeddings give +0.009, and −0.011 on the strict test. That fits some memorizing of places, though the strict test also trains on less data.

Independent check with night lights

Night lights stayed out of the model and serve as a second observation of the outage. NASA Black Marble brightness fell to a median of 51% of normal on nights when under half of a county's customers had power, and to 87% when 50% to 80% had power. At 80% to 95% it stayed at 97%, so the satellite reliably sees only large outages.

The predictions agree with it where night lights respond. In the quarter of pairs with the longest predicted outages, the darkest clear night within 7 days had a median of 70% of normal, against 90% to 92% in the other quarters. The rank correlation between prediction and dimming is 0.35 overall, close to the 0.37 of the observed duration itself. By hurricane it runs from −0.03 to 0.64, and where it is near zero (Florence, Zeta) the observed duration does no better.

Limits

- **Severity of a new storm.** The model ranks counties well but misjudges the overall level. In the worst case, Irma, it predicted 13% of counties over 24 hours and 52% were.
- **Small gains are fragile.** A ridge regression on wind alone scores 0.76, and gains moved by up to 0.006 between two versions of scikit-learn.
- **Nine storms, county scale.** I chose the states for each storm by hand, and EAGLE-I covers only utilities that publish outage maps. A model trained and tested only on pairs that reached 10% scores 0.74.

Next steps

Next come more hurricanes and a model of storm-level severity, forecast wind and rain as inputs, and outcomes finer than the county. The night-light result matters most where no outage records exist, as in my earlier project on Philippine typhoons.

Data, code and sources

Notebooks, result tables and figure code accompany this brief in the project repository. All inputs are public. Outages: Brelsford et al. (2024), *Scientific Data* 11, 271. Embeddings: Brown et al. (2025), arXiv:2507.22291. Other inputs: NOAA RTMA and IBTrACS, PRISM, NLCD, SRTM, American Community Survey, HHS emPOWER, Sentinel-1, MERIT Hydro and NASA Black Marble.